Representation of Switched Systems by Perturbed Control Systems

J. L. Mancilla-Aguilar† R. García†† E. Sontag‡ and Y. Wang‡‡

Abstract— **This work provides representations of switched systems described by controlled differential inclusions, in terms of perturbed control systems whose dynamics are given by differential equations, and whose inputs consist of the original controls together with disturbances that evolve in compact sets. Several applications dealing with properties of stability with respect to inputs and of detectability, and their characterizations in terms of Lyapunov functions, are derived as consequences of the representation theorem for such switched systems, and more generally, for systems with switching signals governed by digraphs.**

I. INTRODUCTION

Consider a switched system described by forced differential inclusions

$$\dot{x}(t) \in F_{\sigma(t)}(x(t), u(t)), \qquad y(t) = h(x(t)), \tag{1}$$

with $\sigma : [0, +\infty) \to \Gamma$ an arbitrary *switching signal* and Γ the index set. In the special case when the set-valued map F_γ ($\gamma \in \Gamma$) is single-point valued (that is, when $F_\gamma(\xi, \mu)$ consists of a single point) for each $\gamma \in \Gamma$, the switched system (1) of differential inclusions reduces to a switched system of differential equations of the following type:

$$\dot{x}(t) = f_{\sigma(t)}(x(t), u(t)), \qquad y(t) = h(x(t)). \tag{2}$$

Various stability properties for such systems were studied and characterized in terms of Lyapunov functions (see [5], [6], [7], [8]). In [7] in particular, it was proved that, under suitable hypotheses, the set of maximal trajectories of system (2) is dense in the set of maximal trajectories of a system with controls and perturbations. By using this fact, different results about perturbed control systems described by differential equations could be extended to switched systems (2) ([7], [8]).

More generally, to take into account some of the uncertainty caused by errors and disturbances that are inevitable in any real-world control problem, it is natural to consider a more general model as in (1).

In the first part of this work we obtain representations of system (1) by means of perturbed control systems described by differential equations, driven by inputs consisting of the controls of the original system and perturbations that evolve in compact sets. As consequences, these representations enable us to extend straightforwardly some stability results, such as direct and converse Lyapunov theorems, previously obtained for perturbed control systems to system (1) with switchings for the stability properties including *input-output-to-state stability* (IOSS), *input-to-output stability* (IOS), and *state independent input-to-output stability* (SIIOS). These stability properties arise within the *input-to-state stability* (ISS) framework as notions that describe the detectability and external stability when the transient behavior due to the initial values are taken into account.

The second part of this work concerns itself with the Lyapunov characterizations of the IOS, SIIOS and IOSS properties of systems as in (1) with switchings governed by a given digraph $H^* \subseteq \Gamma \times \Gamma$. This restriction of the switchings enables us, for example, to model the restrictions imposed by the discrete process of an hybrid system whose continuous portion is (2) (see [1], [6]).

As it can be seen in later sections, the restriction on the switchings leads to a family of subclasses of switching signals, where a switching signal belongs to a certain subclass whenever it takes values in a certain strongly connected component of the digraph H^*. The behavior of a switched system when the switchings are restricted to these classes plays a significant role in this work, since it will enable us to characterize the stability properties of the switched system in terms of these same properties when the switchings are restricted to these subclasses.

Due to the length restriction, we have omitted most of the proofs. For more detailed discussions, proofs and derivations, see [9] and [10].

The paper is organized as follows. In Section II we present the basic notations, the class of switched systems that we address, and the main results on the representation of systems as in (1) by perturbed control systems defined by differential equations. As immediate applications of our results on representations, in Section III we extend previous results on Lyapunov characterizations of input/output stability and detectability properties for systems of differential equations to switched systems defined by differential inclusions. In Section IV we discuss the stability properties for systems as in (1) with switchings governed by a digraph H^*. We present characterizations of the uniform IOS, OLIOS, SIIOS and IOSS stability properties for such systems both in terms of the stability properties of the system with switchings restricted to the subclasses resulted from H^*, and in terms of Lyapunov functions for this restricted system. Finally, the conclusions are given in Section V.

† Department of Mathematics, Faculty of Engineering, University of Buenos Aires. jmancil@fi.uba.ar Work partially supported by UBA I039

†† Department of Physics & Mathematics of the Instituto Tecnológico de Buenos Aires; and the Faculty of Engineering of the University of Buenos Aires. ragarcia@itba.edu.ar Work partially supported by UBA I039

‡ Department of Mathematics, Rutgers University, New Brunswick, NJ. sontag@math.rutgers.edu Work partially supported by US Air Force Grant F49620-01-1-0063 and NSF Grant CCR-0206789.

‡‡ Department of Mathematics, Florida Atlantic University, FL. ywang@math.fau.edu Work partially supported by NSF Grant DMS-0072620 and Chinese National Natural Science Foundation grant 60228003

II. SWITCHED SYSTEMS

In this work, we use $|\cdot|$ to denote the Euclidean norm for any given $\mathbb{R}^q$ and $\mathcal{B}_q$ to denote its closed unit ball. Given a metric space E, we denote by $\mathcal{M}(E)$ the set of Lebesgue measurable functions $\eta : [0,+\infty) \to E$ that are *locally essentially bounded*. In case $E = \mathbb{R}^m$ we write $\mathcal{U}$ instead of $\mathcal{M}(\mathbb{R}^m)$. For $u \in \mathcal{U}$ we denote by $\|u\|$ the (possibly infinite) L^m_∞-norm of u and, for any $t \geq 0$, $\|u\|_{[0,t]}$ stands for the L^m_∞-norm of u restricted to the interval $[0,t]$.

Let X be a metric space. We denote the distance from a point $\xi \in X$ to a set $A \subseteq X$ by $dist(\xi, A)$ and by $d_H(A,B)$ the *Hausdorff distance* between two nonempty closed subsets A and B of X.

We denote $\mathcal{K}(X)$ the set of nonempty compacts subsets of X and we recall that the Hausdorff distance d_H is a metric on $\mathcal{K}(X)$. If X is a normed vector space, with norm $\|\cdot\|$, and $A \subseteq X$, $\mathrm{co}A$ and $\mathrm{cl}A$ stand for the convex hull and the closure of A respectively. We define $\|A\| := \sup\{\|a\| : a \in A\}$.

Given another metric space Z, we say that a set-valued map $G : Z \to \mathcal{K}(X)$ is continuous (Lipschitz) if it is continuous (Lipschitz) when the Hausdorff distance is considered in $\mathcal{K}(X)$. $\mathcal{C}(Z, \mathcal{K}(X))$ denotes the class of continuous maps from Z to $\mathcal{K}(X)$.

Definition 2.1: Given a nonempty set C, we say that a function $g : [0,+\infty) \to C$ is a switching signal if g verifies one of the two following conditions:

1) g is a piecewise constant function (*i.e.* the set of points where the function g has jumps is finite in each compact subinterval of $[0,+\infty)$, and g is constant between jumps) continuous from the right;
2) there exist a sequence $\{t_k : k \in \mathbb{N}_0\}$ with $0 = t_0 < t_1 < \cdots < t_k < \cdots$ and $\lim_{k\to+\infty} t_k = T_g < +\infty$, a sequence $\{c_k \in C : k \in \mathbb{N}_0\}$ with $c_k \neq c_{k+1}$ for all $k \geq 0$ and a point $c^* \in C$ such that $g(t) = c_k$ for all $t_k \leq t < t_{k+1}$ and $g(t) = c^*$ for all $t \geq T_g$.

In what follows, we will denote by $\mathcal{S}(C)$ the family of all C-valued switching signals and with $\mathcal{S}_{pc}(C)$ the family of all C-valued piecewise constant switching signals.

Remark 2.1: The definition of switching signals that we use here is slightly more general than those usually used in the literature, where a switching signal is a piecewise constant function, that is, the class we have called $\mathcal{S}_{pc}(C)$. Switching signals that do not belong to $\mathcal{S}_{pc}(C)$ are usually related with the so-called Zeno-behavior of a hybrid system (see [12]) and due to this reason they will be here referred to as Zeno-switching signals.

Given a family of locally Lipschitz set-valued maps $\mathcal{P} = \{F_\gamma \in \mathcal{C}(\mathbb{R}^n \times \mathbb{R}^m, \mathcal{K}(\mathbb{R}^n)),\ \gamma \in \Gamma\}$, where Γ is an index set and, without lost of generality $F_\gamma \neq F_{\gamma'}$ if $\gamma \neq \gamma'$, we consider the switched system with inputs

$$\dot{x} \in F_\sigma(x,u) \tag{3}$$

where x takes values in $\mathbb{R}^n$, $u \in \mathcal{U}$, and $\sigma \in \mathcal{S}(\Gamma)$.

For an input $u \in \mathcal{U}$ and a switching signal $\sigma \in \mathcal{S}(\Gamma)$, we say that a locally absolutely continuous function $x : \mathcal{I} \to \mathbb{R}^n$, where $\mathcal{I} = [0,T]$ or $[0,T)$ with $0 < T \leq +\infty$, is a *trajectory* of (3) *corresponding* to $u \in \mathcal{U}$ and to $\sigma \in \mathcal{S}(\Gamma)$ if $\dot{x}(t) \in F_{\sigma(t)}(x(t),u(t))$ $a.e.\, t \in \mathcal{I}$.

A trajectory $x : [0,T_x) \to \mathbb{R}^n$ corresponding to $u \in \mathcal{U}$ and to $\sigma \in \mathcal{S}(\Gamma)$ is called *maximal* if it does not have an extension which is a solution corresponding to u and to σ.

For any $\xi \in \mathbb{R}^n$, any $u \in \mathcal{U}$ and any $\sigma \in \mathcal{S}(\Gamma)$, we denote with $\mathcal{T}^s(\xi,u,\sigma)$ the collection of all the maximal trajectories x of (3) corresponding to u and to σ that satisfy $x(0) = \xi$.

A. Main Results on Parameterized Representations

In what follows we will establish one of the main results of this work, which asserts that under suitable hypotheses on the family $\mathcal{P}$ the set of maximal trajectories of system (3) is dense, in a sense that we will make precise, in the set of maximal trajectories of a system with inputs described by the ordinary differential equation

$$\dot{z} = f(z,u,d,\omega), \tag{4}$$

where z takes values in $\mathbb{R}^n$, $u \in \mathcal{U}$, $d \in \mathcal{M}(D)$, with D a compact metric space, and $\omega \in \mathcal{M}(\mathcal{B}_n)$.

To state the main results, we consider for the family $\mathcal{P}$ the following hypotheses:

- **C1** : The family $\mathcal{P}$ is uniformly locally Lipschitz, *i.e.*, for each $N \in \mathbb{N}$, there exists $l_N \geq 0$ such that
$$d_H(F_\gamma(\xi,\mu), F_\gamma(\xi',\mu')) \leq l_N(|\xi-\xi'| + |\mu-\mu'|)$$
for all $(\xi,\mu),(\xi',\mu') \in N\mathcal{B}_n \times N\mathcal{B}_m$ and all $\gamma \in \Gamma$.
- **C2** : The family $\mathcal{P}$ is pointwise equibounded, *i.e.*, for each $(\xi,\mu) \in \mathbb{R}^n \times \mathbb{R}^m$ there exists $M_{(\xi,\mu)} \geq 0$ such that $\|F_\gamma(\xi,\mu)\| \leq M_{(\xi,\mu)}$ for all $\gamma \in \Gamma$.

Remark 2.2: Hypotheses **C1** and **C2** are trivially satisfied when Γ is a finite set.

Theorem 2.1: Suppose that $\mathcal{P}$ verifies **C1** and **C2**. Then there exist a compact metric space D, an injective function $\iota : \Gamma \to D$ and a continuous function $f : \mathbb{R}^n \times \mathbb{R}^m \times D \times \mathcal{B}_n \to \mathbb{R}^n$ such that:

1) $\iota(\Gamma)$ is dense in D.
2) $f(\cdot,\cdot,\nu,\cdot)$ is locally Lipschitz uniformly on $\nu \in D$, *i.e.*, for each compact subset K of $\mathbb{R}^n \times \mathbb{R}^m$ there is some constant c_K so that $|f(\xi,\mu,\nu,\zeta) - f(\xi',\mu',\nu,\zeta')| \leq c_K(|\xi-\xi'|+|\mu-\mu'|+|\zeta-\zeta'|)$, for all $(\xi,\mu),(\xi',\mu') \in K$, all $\zeta,\zeta' \in \mathcal{B}_n$ and all $\nu \in D$.
3) For each $\xi \in \mathbb{R}^n$, $\mu \in \mathbb{R}^m$ and $\gamma \in \Gamma$,
$$\mathrm{co}\, F_\gamma(\xi,\mu) = \{f(\xi,\mu,\iota(\gamma),\zeta) : \zeta \in \mathcal{B}_n\}.$$
4) Given $u \in \mathcal{U}$, $\sigma \in \mathcal{S}(\Gamma)$ and a maximal trajectory x of (3) corresponding to u and σ, there exists $\omega \in \mathcal{M}(\mathcal{B}_n)$ such that x is a maximal trajectory of (4) corresponding to u, $d_\sigma = \iota \circ \sigma$ and ω.
5) The trajectories of (3) are dense in the set of trajectories of (4) in the two following senses:
   (a) Given $\varepsilon > 0$ and a trajectory $z : [0,T] \to \mathbb{R}^n$ of (4) corresponding to $u \in \mathcal{U}$, $d \in \mathcal{M}(D)$ and $\omega \in$

$\mathcal{M}(\mathcal{B}_n)$, there exist $\sigma \in \mathcal{S}_{pc}(\Gamma)$ and a trajectory x of (3) corresponding to u and σ such that $x(0) = z(0)$ and $|z(t) - x(t)| < \varepsilon \quad \forall t \in [0, T]$.

(b) Given a trajectory $z : [0, T) \to \mathbb{R}^n$ of (4), with $T < +\infty$ ($T = +\infty$), corresponding to $u \in \mathcal{U}$, $d \in \mathcal{M}(D)$ and $\omega \in \mathcal{M}(\mathcal{B}_n)$ and a continuous function $r : [0, T) \to \mathbb{R}_{>0}$, there exist $\sigma \in \mathcal{S}(\Gamma)$ ($\sigma \in \mathcal{S}_{pc}(\Gamma)$) and a trajectory x of (3) corresponding to u and σ such that $|z(t) - x(t)| < r(t) \quad \forall t \in [0, T)$.

Remark 2.3: Parts 1. to 3. of Theorem 2.1 contain as a particular case a result obtained in [8] for switched systems described by differential equations (Theorem 3.2 of [8]).

Remark 2.4: Part 5.(a) of Theorem 2.1 asserts that for fixed $T > 0$, $u \in \mathcal{U}$ and $\xi \in \mathbb{R}^n$, the set of trajectories of (3) corresponding to u and to every $\sigma \in \mathcal{S}_{pc}(\Gamma)$ with starting point ξ and defined on $[0, T]$, are dense in the set of trajectories of (4) corresponding to u, to every $d \in \mathcal{M}(D)$ and to every $\omega \in \mathcal{M}(\mathcal{B}_n)$ with the same starting point ξ and defined on $[0, T]$, when this last set of trajectories is equipped with the topology of uniform convergence on the interval $[0, T]$.

On the other hand, part 5.(b) states that for fixed $0 < T < +\infty$ ($T = +\infty$) and $u \in \mathcal{U}$, the set of trajectories of (3) corresponding to u and to every $\sigma \in \mathcal{S}(\Gamma)$ ($\sigma \in \mathcal{S}_{pc}(\Gamma)$) that are defined on $[0, T)$, are dense in the the set of trajectories of (4) corresponding to u, to every $d \in \mathcal{M}(D)$ and to every $\omega \in \mathcal{M}(\mathcal{B}_n)$ that are defined on $[0, T)$, when in the last set we consider the $\mathcal{C}^0$ Whitney topology.

We point out that part 5.(b) does not hold if we consider only piecewise constant switching signals. That is the reason why we must also consider Zeno-switching signals.

As a corollary of Theorem 2.1 and some results on forced differential inclusions, we also obtain the following representation theorem

Theorem 2.2: Suppose that $\mathcal{P}$ verifies **C1** and **C2**. Then there exists a locally Lipschitz function $f^* : \mathbb{R}^n \times \mathbb{R}^m \times \mathcal{B}_n \to \mathbb{R}^n$ such that:

1) for each $\xi \in \mathbb{R}^n$ and $\mu \in \mathbb{R}^m$,

$$\operatorname{co}\operatorname{cl}\left(\bigcup_{\gamma \in \Gamma} F_\gamma(\xi, \mu)\right) = \{f^*(\xi, \mu, \zeta) : \zeta \in \mathcal{B}_n\}.$$

2) Given $u \in \mathcal{U}$, $\sigma \in \mathcal{S}(\Gamma)$ and a maximal trajectory x of (3) corresponding to u and σ, there exists $\omega \in \mathcal{M}(\mathcal{B}_n)$ such that x is a maximal trajectory of

$$\dot{x} = f^*(x, u, \omega). \tag{5}$$

3) The trajectories of (3) are dense in the set of trajectories of (5) in the two following senses:

(a) Given $\varepsilon > 0$ and a trajectory $z : [0, T] \to \mathbb{R}^n$ of (5) corresponding to $u \in \mathcal{U}$ and $\omega \in \mathcal{M}(\mathcal{B}_n)$ there exist $\sigma \in \mathcal{S}_{pc}(\Gamma)$ and a trajectory x of (3) corresponding to u and σ such that $x(0) = z(0)$ and $|z(t) - x(t)| < \varepsilon \quad \forall t \in [0, T]$.

(b) Given a trajectory $z : [0, T) \to \mathbb{R}^n$ of (5), with $T < +\infty$ ($T = +\infty$), corresponding to $u \in \mathcal{U}$ and $\omega \in \mathcal{M}(\mathcal{B}_n)$ and a continuous function $r : [0, T) \to \mathbb{R}_{>0}$, there exist $\sigma \in \mathcal{S}(\Gamma)$ ($\sigma \in \mathcal{S}_{pc}(\Gamma)$) and a trajectory x of (3) corresponding to u and σ such that $|z(t) - x(t)| < r(t) \quad \forall t \in [0, T)$.

The two results complement each other: while Theorem 2.1 provides a representation of the given switched differential inclusion in terms of a switched system of differential equations (driven by the same switching signal, provided that we identify σ with $\iota \circ \sigma$), Theorem 2.2 is more abstract, and it provides a representation in terms of a non-switched system of differential equations. This latter form, which looses the information about the switching signal, is of use for theoretical purposes, since it allows reducing questions about switched differential inclusions into questions about ordinary differential equations; example of its use appear later in the paper.

B. A Relaxation Theorem

The proofs of Theorems 2.1 and 2.2 rely on some results about parameterizations of set valued maps and on the following result which is also important by itself.

Let us consider the time-varying system with inputs described by

$$\dot{x} \in G(t, x, d) \tag{6}$$

where $t \geq 0$, x takes values in $\mathbb{R}^n$ and $d \in \mathcal{M}(D)$ with D a separable metric space. We also consider the relaxation of (6)

$$\dot{x} \in \operatorname{co} G(t, x, d). \tag{7}$$

Theorem 2.3: Suppose that the set-valued map $G : \mathbb{R}_{\geq 0} \times \mathbb{R}^n \times D \to \mathcal{K}(\mathbb{R}^n)$, with D a separable metric space, satisfies the following hypotheses:

$\mathbf{H_1}$ $G(\cdot, \xi, \nu)$ is measurable for each $(\xi, \nu) \in \mathbb{R}^n \times D$;

$\mathbf{H_2}$ $G(t, \cdot, \cdot)$ is continuous for each $t \geq 0$;

$\mathbf{H_3}$ for each compact subset $K \subset \mathbb{R}^n \times D$ there exists $L_K \in L^1_{loc}(\mathbb{R}_{\geq 0}, \mathbb{R})$ such that for each $(\xi, \nu), (\xi', \nu) \in K$,

$$d_H(G(t, \xi, \nu), G(t, \xi', \nu)) \leq L_K(t)|\xi - \xi'| \quad \forall t \geq 0;$$

$\mathbf{H_4}$ for each compact subset $K \subset \mathbb{R}^n \times D$ there exists $\alpha_K \in L^1_{loc}(\mathbb{R}_{\geq 0}, \mathbb{R})$ such that for each $(\xi, \nu) \in K$

$$\|G(t, \xi, \nu)\| \leq \alpha_K(t) \quad \forall t \geq 0,$$

and let D' be a dense subset of D.

Then, given a maximal trajectory $z : [0, T_z) \to \mathbb{R}^n$ of (7) corresponding to $d^* \in \mathcal{M}(D)$ the following hold:

1) Given $0 < T < T_z$ and $\varepsilon > 0$, there exist a piecewise constant switching signal $d \in \mathcal{S}_{pc}(D')$ and a maximal trajectory x of (6) corresponding to d such that $x(0) = z(0)$ and

$$|x(t) - z(t)| < \varepsilon \quad \forall t \in [0, T]; \tag{8}$$

2) Given $0 < T \leq T_z$ and a continuous function $r : [0, T) \to \mathbb{R}_{>0}$, there exist a switching signal $d \in \mathcal{S}(D')$ ($d \in \mathcal{S}_{pc}(D')$ if $T_z = +\infty$) and a maximal trajectory x of (6) corresponding to d such that

$$|x(t) - z(t)| < r(t) \quad \forall t \in [0, T). \tag{9}$$

Remark 2.5: Theorem 2.3 remains valid, if we consider that the domain of the first variable of G is a finite interval $\mathcal{I}$ of the form $[0, a]$ or $[0, a)$, and we replace $\mathbb{R}_{\geq 0}$ by $\mathcal{I}$ in hypotheses $(H_1) - (H_4)$.

Remark 2.6: Parts 1. and 2. of Theorem 2.3 can be considered as extensions of Filippov-Ważewski Theorem and Theorem 1 of [2], respectively, since they are particular cases of Theorem 2.3. To see it, just consider that D in Theorem 2.3 is a singleton.

The following result on forward completeness of control systems

$$\dot{x} = f(x, u) \tag{10}$$

where x takes values in $\mathbb{R}^n$, $u \in \mathcal{M}(D)$, with D a separable metric space and $f : \mathbb{R}^n \times D \to \mathbb{R}^n$ continuous and locally Lipschitz in x uniformly in compact subsets of D, follows readily from Theorem 2.3:

Theorem 2.4: Let D' be a dense subset of D. Then system (10) is forward complete if and only if it is forward complete with respect to $\mathcal{S}(D')$. ■

III. STABILITY OF SWITCHED SYSTEMS

In this section we discuss several output stability properties of a system as in (3) with an output map $y = h(x)$. In what follows, we will restrict our consideration of switching signals to the class $\mathcal{S}_{pc}(\Gamma)$. For simplicity, we will use $\mathcal{S}_{pc}$ for $\mathcal{S}_{pc}(\Gamma)$.

We will say that the system (3) is *forward complete* with respect to a subclass of switching signals $\mathcal{S}^*_{pc} \subseteq \mathcal{S}_{pc}$ if every maximal solution $x \in \mathcal{T}^s(\xi, u, \sigma)$ is defined for all $t \geq 0$. We just say that system (3) is *forward complete* when it is forward complete with respect to $\mathcal{S}_{pc}$. It follows that (3) is forward complete with respect to $\mathcal{S}_{pc}$ if and only if $\dot{x} \in F_\gamma(x, u)$ is forward complete for each $\gamma \in \Gamma$.

Definition 3.1: Given a subclass of switched signals $\mathcal{S}^*_{pc} \subseteq \mathcal{S}_{pc}$, the system (3) is

∘ *uniformly input-output-to-state stable with respect to* $\mathcal{S}^*_{pc}$ (UIOSS w.r.t. $\mathcal{S}^*_{pc}$) if there exist some $\beta \in \mathcal{KL}$, $\alpha \in \mathcal{K}$ and $\theta \in \mathcal{K}$ such that for all $\xi \in \mathbb{R}^n$, all $u \in \mathcal{U}$, all $\sigma \in \mathcal{S}^*_{pc}$ and all $x \in \mathcal{T}^s(\xi, u, \sigma)$,

$$|x(t)| \leq \beta(|\xi|, t) + \theta(\|y\|_{[0,t]}) + \alpha(\|u\|_{[0,t]}) \tag{11}$$

for all $t \in [0, T_x)$;

∘ *uniformly input-to-output stable with respect to* $\mathcal{S}^*_{pc}$(UIOS w.r.t. $\mathcal{S}^*_{pc}$), if it is forward complete w.r.t $\mathcal{S}^*_{pc}$ and there exist $\beta \in \mathcal{KL}$, $\gamma \in \mathcal{K}$ such that for all $\xi \in \mathbb{R}^n$, all $u \in \mathcal{U}$, all $\sigma \in \mathcal{S}^*_{pc}$ and all $x \in \mathcal{T}^s(\xi, u, \sigma)$,

$$|y(t)| \leq \beta(|\xi|, t) + \gamma(\|u\|) \quad \forall t \geq 0; \tag{12}$$

∘ *uniformly output-Lagrange input-to-output stable with respect to* $\mathcal{S}^*_{pc}$ (UOLIOS w.r.t. $\mathcal{S}^*_{pc}$) if it is UIOS w.r.t. $\mathcal{S}^*_{pc}$ and there exist $\mathcal{K}$-functions λ_1 and λ_2 such that for all $\xi \in \mathbb{R}^n$, all $u \in \mathcal{U}$, all $\sigma \in \mathcal{S}^*_{pc}$ and all $x \in \mathcal{T}^s(\xi, u, \sigma)$,

$$|y(t)| \leq \max\{\lambda_1(|h(\xi)|), \lambda_2(\|u\|)\} \quad \forall t \geq 0; \tag{13}$$

∘ *uniformly state independent input-to-output stable with respect to* $\mathcal{S}^*_{pc}$ (USIIOS w.r.t. $\mathcal{S}^*_{pc}$) if it is forward complete w.r.t $\mathcal{S}^*_{pc}$ and there exist $\beta \in \mathcal{KL}$ and $\alpha \in \mathcal{K}$ such that for all $\xi \in \mathbb{R}^n$, all $u \in \mathcal{U}$, all $\sigma \in \mathcal{S}^*_{pc}$ and all $x \in \mathcal{T}^s(\xi, u, \sigma)$,

$$|y(t)| \leq \beta(|h(\xi)|, t) + \alpha(\|u\|_{[0,t]}) \quad \forall t \geq 0. \tag{14}$$

When $\mathcal{S}^*_{pc} = \mathcal{S}_{pc}$, we will mention the properties above without making reference to the switching class involved.

Remark 3.1: We note that the stability properties just introduced are natural extensions of those given in [3] and [11] for systems described by differential equations.

Various stability properties are particular cases of these ones. For example, for systems without inputs, the globally asymptotic stability of the system with respect to a forward invariant closed set $\mathcal{A}$ is equivalent to the SIIOS property, if we consider as output map $h(\xi) = dist(\xi, \mathcal{A})$. The well-known input-to-state stability property (ISS) is also a particular case of these properties. In fact, if we consider as output map $h(\xi) = 0$, then the IOSS property becomes the ISS property. On the other hand, if we consider $h(\xi) = \xi$, then both the IOS and SIIOS are equivalent to ISS.

In order to establish the main result of this section we introduce the following stability notion. Given a subclass of switching signals $\mathcal{S}^*_{pc} \subseteq \mathcal{S}_{pc}$, we say that system (3) is *uniformly (with respect to* $\mathcal{S}^*_{pc}$*) bounded input bounded state stable* (UBIBS w.r.t $\mathcal{S}^*_{pc}$) if there exists a nondecreasing function $\varphi : \mathbb{R}_{\geq 0} \to \mathbb{R}_{\geq 0}$ such that the following holds:

$$|x(t)| \leq \max\{\varphi(|\xi|), \varphi(\|u\|)\} \quad \forall\, t \geq 0, \tag{15}$$

for all $\xi \in \mathbb{R}^n$, all $u \in \mathcal{U}$, all $s \in \mathcal{S}^*_{pc}$ and all $x \in \mathcal{T}^s(\xi, u, s)$. When $\mathcal{S}^*_{pc} = \mathcal{S}_{pc}$ we just say that (3) is UBIBS.

A. Lyapunov Characterizations

In this section we characterize the UIOSS, UIOS, USIIOS and UOLIOS w.r.t. $\mathcal{S}_{pc}$ properties in terms of Lyapunov functions. With this aim, we introduce the following definitions.

Definition 3.2: Given a family of locally Lipschitz set-valued maps $\mathcal{P} = \{F_\gamma : \mathbb{R}^n \times \mathbb{R}^m \to \mathcal{K}(\mathbb{R}^n) : \gamma \in \Gamma\}$ and an output-map h, a $\mathcal{C}^1$ function $V : \mathbb{R}^n \to \mathbb{R}$ is

1) a *common* IOSS-*Lyapunov function for* $(\mathcal{P}, h)$ if it is positive definite, radially unbounded and there exist functions ρ, χ and ν of class $\mathcal{K}_\infty$ that verify, $\forall \xi \in \mathbb{R}^n, \forall \mu \in \mathbb{R}^m, \forall \gamma \in \Gamma, \forall v \in F_\gamma(\xi, \mu)$

$$DV(\xi)v \leq -\rho(|\xi|) + \chi(|h(\xi)|) + \nu(|\mu|);$$

2) a *common* IOS-*Lyapunov function for* $(\mathcal{P}, h)$ if
   (a) there exist two $\mathcal{K}_\infty$-functions α_1 and α_2 such that $\alpha_1(|h(\xi)|) \leq V(\xi) \leq \alpha_2(|\xi|) \quad \forall\, \xi \in \mathbb{R}^n$;
   (b) there exist $\alpha \in \mathcal{KL}$ and $\chi \in \mathcal{K}$ so that, $\forall \xi \in \mathbb{R}^n$, $\forall \mu \in \mathbb{R}^m$, $\forall \gamma \in \Gamma$ and $\forall v \in F_\gamma(\xi, \mu)$,

$$V(\xi) \geq \chi(|\mu|) \Rightarrow DV(\xi)v \leq -\alpha(V(\xi), |\xi|);$$

3) a *common* SIIOS-*Lyapunov function for* $(\mathcal{P}, h)$ if
   (a) there exist two $\mathcal{K}_\infty$-functions α_1 and α_2 such that $\alpha_1(|h(\xi)|) \leq V(\xi) \leq \alpha_2(|h(\xi)|) \quad \forall\, \xi \in \mathbb{R}^n$;
   (b) there exist two $\mathcal{K}_\infty$-functions χ and ρ so that, $\forall \xi \in \mathbb{R}^n$, $\forall \mu \in \mathbb{R}^m$, $\forall \gamma \in \Gamma$ and $\forall v \in F_\gamma(\xi, \mu)$,
$$V(\xi) \geq \chi(|\mu|) \Rightarrow DV(\xi)v \leq -\rho(V(\xi));$$
4) a *common* OLIOS-*Lyapunov function for* $(\mathcal{P}, h)$ if 3-(a) holds and there exist $\chi \in \mathcal{K}_\infty$ and $\alpha_3 \in \mathcal{KL}$ so that, $\forall \xi \in \mathbb{R}^n$, $\forall \mu \in \mathbb{R}^m$, $\forall \gamma \in \Gamma$ and $\forall v \in F_\gamma(\xi, \mu)$,
$$V(\xi) \geq \chi(|\mu|) \Rightarrow DV(\xi)v \leq -\alpha_3(V(\xi), |\xi|) \quad \blacksquare$$

Remark 3.2: It can be easily shown, with arguments similar to those used in [3], that the existence of a common IOSS-Lyapunov function V for $(\mathcal{P}, h)$ assures that the system (3) is UIOSS w.r.t $\mathcal{S}_{pc}(\Gamma)$.

On the other hand, by using arguments similar to those used in [11], and with the additional hypothesis that system (3) is forward complete w.r.t. $\mathcal{S}_{pc}$, it can be proven that system (3) is USIIOS w.r.t $\mathcal{S}_{pc}$ if there exists a SIIOS-Lyapunov function V for $(\mathcal{P}, h)$. Finally, under the assumption that (3) is UBIBS with respect to $\mathcal{S}_{pc}$, it can be proved, by using also arguments similar to those used in [11], that the existence of a common IOS-Lyapunov (resp. OLIOS-Lyapunov) function V for $(\mathcal{P}, h)$ assures that the system (3) is UIOS (resp. UOLIOS) w.r.t $\mathcal{S}_{pc}$. (Note that UBIBS property implies the forward completeness of (3)).

The following result on the existence of common Lyapunov functions for switched systems given by differential inclusions is the converse of the results pointed in Remark 3.2. The existence of such a function in the IOSS case can be obtained by combining Theorem 2.2 and Theorem 1 of [3]. In the case of IOS and related stability properties, the existence of those functions follow from Theorem 2.2 and a straightforward generalization of the Lyapunov results in [11] to systems described by differential equations subject to disturbances taking values in compact sets. (The proof of these facts can be found in Theorems 6–8 of [9]).

Theorem 3.1: Suppose that the collection $\mathcal{P}$ satifies assumptions **C1** and **C2**, and that h is locally Lipschitz. Then the following hold:

1) If system (3) is UIOSS w.r.t. $\mathcal{S}_{pc}$ and $F_\gamma(0, 0) = \{0\}$ for all $\gamma \in \Gamma$, then there exists a $\mathcal{C}^\infty$ common IOSS-Lyapunov function V for $(\mathcal{P}, h)$.
   Suppose now that system (3) is UBIBS w.r.t. $\mathcal{S}_{pc}$.
2) If system (3) is UIOS (resp. UOLIOS, USIIOS) w.r.t. $\mathcal{S}_{pc}$ then there exists a $\mathcal{C}^\infty$ common IOS (resp.OLIOS, SIIOS) Lyapunov function V for $(\mathcal{P}, h)$.

Remark 3.3: As in the work [11], the UBIBS condition is not needed in the case of USIIOS, that is, the result of part 2) of Theorem 3.1 referring to USIIOS holds true for systems that are foward complete w.r.t. $\mathcal{S}(\Gamma)$ but are not necessarily UBIBS. We must remark that in this case we have to assume that (3) is forward complete w.r.t $\mathcal{S}(\Gamma)$ since the forward completeness property with respect to merely $\mathcal{S}_{pc}$ is not enough to guarantee the forward completeness property of the parameterization of (3) as given in Theorem 2.2.

IV. Systems with switching governed by a graph

Below we study some stability properties of switched systems whose switchings are governed by a given digraph $H^* \subseteq \Gamma \times \Gamma$, more precisely,

Definition 4.1: We say that a switching signal $\sigma \in \mathcal{S}_{pc}$ with a sequence of switching times $\{t_i,\, i \in \mathbb{N}_0\}$ is an *admissible switching signal* if and only if $(\sigma(t_i), \sigma(t_{i+1})) \in H^*$ for all i.

We denote by $\mathcal{S}_{pc}^{\mathrm{adm}}$ the class of all admissible switching signals.

Remark 4.1: Observe that if we consider the set-valued map $H : \Gamma \to 2^\Gamma$, defined by $H(\gamma) = \{\gamma' : \ (\gamma, \gamma') \in H^*\}$, then we have that $\sigma \in \mathcal{S}_{pc}^{adm}$ if and only if $\sigma(t_{i+1}) \in H(\sigma(t_i))$ for all i.

In what follows we obtain, under suitable hypotheses, characterizations of UIOSS, USIIOS and UOLIOS of system (3) with switching signals constrained to $\mathcal{S}_{pc}^{\mathrm{adm}}$. With this aim we will use some terminology and results from graph theory (see [4] for details).

Given γ, $\gamma' \in \Gamma$, we say that γ' is *accessible* from γ if there exists a finite sequence $\gamma = \gamma_0, \ldots, \gamma_k = \gamma'$ with $k \in \mathbb{N}_0$ such that $(\gamma_i, \gamma_{i+1}) \in H^*$ for all $0 \leq i < k$. In other words, $\gamma_{i+1} \in H(\gamma_i)$ for all $0 \leq i < k$. Then, if we consider in Γ the relation defined by: $\gamma \sim \gamma'$ if and only if γ is accessible from γ' and γ' is accessible from γ, it follows that this is an equivalence relation and that its equivalence classes are the so-called strongly connected components of the digraph H^*.

Assumption **C3**. H^* has a finite number of strongly connected components $\Gamma_1, \ldots, \Gamma_N$.

Theorem 4.1: Suppose that Assumptions **C1**, **C2** and **C3** hold. Then system (3) is

1) UIOSS w.r.t. $\mathcal{S}_{pc}^{\mathrm{adm}}(\Gamma)$ if and only if it is UIOSS w.r.t. $\mathcal{S}_{pc}(\Gamma_i)$ for each $i \in \{1, \ldots, N\}$;
2) USIIOS w.r.t. $\mathcal{S}_{pc}^{\mathrm{adm}}(\Gamma)$ if and only if it is USIIOS w.r.t. $\mathcal{S}_{pc}(\Gamma_i)$ for each $i \in \{1, \ldots, N\}$;
3) UOLIOS and UBIBS w.r.t. $\mathcal{S}_{pc}^{\mathrm{adm}}(\Gamma)$ if and only if it is UOLIOS and UBIBS w.r.t. $\mathcal{S}_{pc}(\Gamma_i)$ for each $i \in \{1, \ldots, N\}$. $\blacksquare$

A. Some remarks about the UIOS case

Unfortunately, there is no characterization for the UIOS w.r.t. $\mathcal{S}_{pc}^{\mathrm{adm}}$ property analogous to those stated in Theorem 4.1 for the other stability properties, even in the case of a UBIBS system.

It is not hard to prove by some approximation arguments that if a system is UIOS w.r.t. $\mathcal{S}_{pc}^{\mathrm{adm}}$ then it is UIOS w.r.t. $\mathcal{S}_{pc}(\Gamma^*)$ for each strongly connected component Γ^* of H^*.

But the converse fails even when the system satisfies the additional hypothesis of being UBIBS stable. For a more detailed discussions with a counterexample, see [10].

The following sufficient condition for (3) to be UIOS w.r.t. $\mathcal{S}_{pc}^{\mathrm{adm}}$ is a simple consequence of Theorem 4.1.

Proposition 4.1: Suppose that Assumptions **C1**, **C2** and **C3** hold. Then, system (3) is UIOS w.r.t. $\mathcal{S}_{pc}^{\mathrm{adm}}$ if it is UBIBS w.r.t. $\mathcal{S}_{pc}(\Gamma_i)$ for $i = 1,\ldots,N$, and there exists a continuous function $h_0 : \mathbb{R}^n \to \mathbb{R}_{\geq 0}$ that verifies the following:

1) there exists $\chi \in \mathcal{K}_\infty$ such that $|h(\xi)| \leq \chi(h_0(\xi))$ for all $\xi \in \mathbb{R}^n$ and
2) system (3) with output $y = h_0(x)$ is UOLIOS w.r.t. $\mathcal{S}_{pc}(\Gamma_i)$ for each $i \in \{1,\ldots,N\}$.

B. Lyapunov Characterizations

In this section we obtain, for graph governed switched systems, some results similar to those of a previous section. In fact, we characterize the UIOSS, USIIOS and UOLIOS w.r.t. $\mathcal{S}_{pc}^{adm}$ properties in terms of Lyapunov functions.

Thus, from remark 3.2, Theorem 4.1 and Proposition 4.1 we easily deduce the following

Theorem 4.2: Suppose that Assumptions **C1**, **C2** and **C3** hold, and let $\mathcal{P}_i = \{F_\gamma : \mathbb{R}^n \times \mathbb{R}^m \to \mathcal{K}(\mathbb{R}^n) : \ \gamma \in \Gamma_i\}$ for $i = 1,\ldots,N$. Then

1) System (3) is UIOSS w.r.t. $\mathcal{S}_{pc}^{\mathrm{adm}}$ if there exists a common IOSS-Lyapunov function V_i for each $(\mathcal{P}_i, h)$ with $i = 1,\ldots,N$.
2) System (3) is USIIOS w.r.t. $\mathcal{S}_{pc}^{\mathrm{adm}}$ if it is forward complete and there exists a common SIIOS-Lyapunov function V_i for each $(\mathcal{P}_i, h)$ with $i = 1,\ldots,N$.
3) System (3) is UOLIOS w.r.t. $\mathcal{S}_{pc}^{\mathrm{adm}}$ if it is UBIBS w.r.t. $\mathcal{S}_{pc}(\Gamma_i)$ for $i = 1,\ldots,N$ and there exists a common OLIOS-Lyapunov function V_i for each $(\mathcal{P}_i, h)$ with $i = 1,\ldots,N$.
4) System (3) is UIOS w.r.t. $\mathcal{S}_{pc}^{\mathrm{adm}}$ if it is UBIBS w.r.t. $\mathcal{S}_{pc}(\Gamma_i)$ for $i = 1,\ldots,N$ and there exists a continuous function $h_0 : \mathbb{R}^n \to \mathbb{R}_{\geq 0}$ that verifies the following:
   - there exists a function χ of class $\mathcal{K}_\infty$ such that $|h(\xi)| \leq \chi(h_0(\xi))$ for all $\xi \in \mathbb{R}^n$;
   - there exists a common OLIOS-Lyapunov function V_i for each $(\mathcal{P}_i, h_0)$ with $i = 1,\ldots,N$.

From Theorem 3.1 it follows that under some additional hypotheses, the converse of Theorem 4.2 holds.

Theorem 4.3: Suppose that Assumptions **C1**, **C2** and **C3** hold and that h is locally Lipschitz. Then the following hold:

1) If system (3) is UIOSS w.r.t. $\mathcal{S}_{pc}^{\mathrm{adm}}$ and $F_\gamma(0,0) = \{0\}$ for all $\gamma \in \Gamma$, then there exists a $\mathcal{C}^\infty$ common IOSS-Lyapunov function V_i for each $(\mathcal{P}_i, h)$ with $i = 1,\ldots,N$.
2) If system (3) is UOLIOS (resp. USIIOS) and UBIBS w.r.t. $\mathcal{S}_{pc}^{\mathrm{adm}}$, then there exists a $\mathcal{C}^\infty$ common OLIOS (resp. SIIOS) Lyapunov function V_i for each $(\mathcal{P}_i, h)$ with $i = 1,\ldots,N$.

It can be seen that the IOSS property reduces to the ISS property in the special case when $h \equiv 0$. Namely, a system as in (3) is *uniformly input-to-state stable* (UISS) w.r.t. a subclass $\mathcal{S}_{pc}^* \subseteq \mathcal{S}_{pc}$ of switched signals if there exist some $\beta \in \mathcal{KL}$ and $\alpha \in \mathcal{K}$ such that for all $\xi \in \mathbb{R}^n$, all $u \in \mathcal{U}$, all $\sigma \in \mathcal{S}_{pc}^*$ and all $x \in \mathcal{T}^s(\xi, u, \sigma)$,

$$|x(t)| \leq \beta(|\xi|, t) + \alpha(\|u\|_{[0,t]}) \quad \forall\, t \geq 0. \qquad (16)$$

Similarly, a $\mathcal{C}^1$ function $V : \mathbb{R}^n \to \mathbb{R}$ is a common ISS-Lyapunov function for a family $\mathcal{P} = \{F_\gamma : \mathbb{R}^n \times \mathbb{R}^m \to \mathcal{K}(\mathbb{R}^n) : \ \gamma \in \Gamma\}$ of locally Lipschitz set valued maps if it is a common IOSS-Lyapunnov function for $(\mathcal{P}, \mathbf{0})$. As a corollary of Theorems 4.2–4.3, one has the following:

Corolary 4.1: Suppose that Assumptions **C1**, **C2** and **C3** hold for system (3) with $\mathcal{S}_{pc}^{\mathrm{adm}}$, and let $\mathcal{P}_i$ be as in Theorem 4.2. Then the system is UISS w.r.t. $\mathcal{S}_{pc}^{\mathrm{adm}}$ if and only if there exists a $\mathcal{C}^\infty$ common ISS-Lyapunov function for $\mathcal{P}_i$ with $i = 1,\ldots N$. ∎

V. CONCLUSIONS

This paper consists of two parts. In the first part we have studied the representation of switched systems of differential inclusions by perturbed control systems defined by differential equations, whose inputs are the original control and perturbations that take values in compact sets.

In the second part we have studied different types of uniform stability properties of switched systems whose subsystems are described by differential inclusions and obtained characterizations of those stability properties in term of common Lyapunov-like functions. For systems with switchings governed by a graph, we have showed that the stability properties can be converted to the stability properties for the same system with switchings in the subclasses resulted from the graph H^*.

REFERENCES

[1] M. Branicky, Multiple Lyapunov functions and other analysis tools for switched and hybrid systems, *IEEE Trans. Automat. Contr.* **43**, (1998) 475-482.

[2] B. Ingalls, E.D. Sontag and Y. Wang, An infinite-time relaxation theorem for differential inclusions, *Proc. Amer. Math. Soc.*, **131**, (2002) 487-499.

[3] M. Krichman, E.D. Sontag and Y. Wang, Input-output-to-state stability, *SIAM J. Control Optim.* **39** (2001) 1874-1928.

[4] D. Jungnickel, *Graphs, Networks and Algorithms*, Springer-Verlag, Berlin, 2003.

[5] D. Liberzon and A.S. Morse, Basic problems in stability and design of switched systems, *IEEE Contr. Syst. Mag.*, **19** (1999) 59-70.

[6] D. Liberzon, *Switching in Systems and Control*, Birkhäuser, Boston, 2003.

[7] J. L. Mancilla-Aguilar and R. A. García, A converse Lyapunov theorem for nonlinear switched systems, *Systems Contr. Lett.* **41**, (2000) 67-71.

[8] J. L. Mancilla-Aguilar and R. A. García, On converse Lyapunov theorems for ISS and iISS switched nonlinear nonlinear systems, *Systems Contr. Lett.* (2001) **42** 47-53.

[9] J.L. Mancilla-Aguilar, R. A. García, E. D. Sontag and Y. Wang, On the representation of switched systems with inputs by perturbed control systems, submitted.

[10] J.L. Mancilla-Aguilar, R. A. García, E. D. Sontag and Y. Wang, Uniform stability properties of switched systems with inputs and outputs, in preparation.

[11] E.D. Sontag and Y. Wang, Lyapunov characterizations of input to output stability, *SIAM J. Control Optim.* **39** (2001) 226-249.

[12] J. Zhang, K.H. Johansson, J. Lygeros and S. Sastry, Zeno hybrid systems, *Int. J. Robust Nonlinear Control* **11** (2001) 435-451.